Analyze Electrodermal Activity (EDA)

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This example can be referenced by citing the package.

This example shows how to use NeuroKit2 to extract the features from **Electrodermal Activity (EDA)**.

```
# Load the NeuroKit package
import neurokit2 as nk
```

Extract the cleaned EDA signal

In this example, we will use a simulated EDA signal. However, you can use any signal you have generated (for instance, extracted from the dataframe using read_acqknowledge().

```
# Simulate 10 seconds of EDA Signal (recorded at 250 samples / second)
eda_signal = nk.eda_simulate(duration=10, sampling_rate=250, scr_number=3, drift=0.
```

Once you have a raw EDA signal in the shape of a vector (i.e., a one-dimensional array), or a list, you can use eda_process() to process it.

```
# Process the raw EDA signal
signals, info = nk.eda_process(eda_signal, sampling_rate=250)
```

Note: It is critical that you specify the correct sampling rate of your signal throughout many processing functions, as this allows NeuroKit to have a time reference.

This function outputs two elements, a *dataframe* containing the different signals (e.g., the raw signal, clean signal, SCR samples marking the different features etc.), and a *dictionary* containing information about the Skin Conductance Response (SCR) peaks (e.g., onsets, peak amplitude etc.).

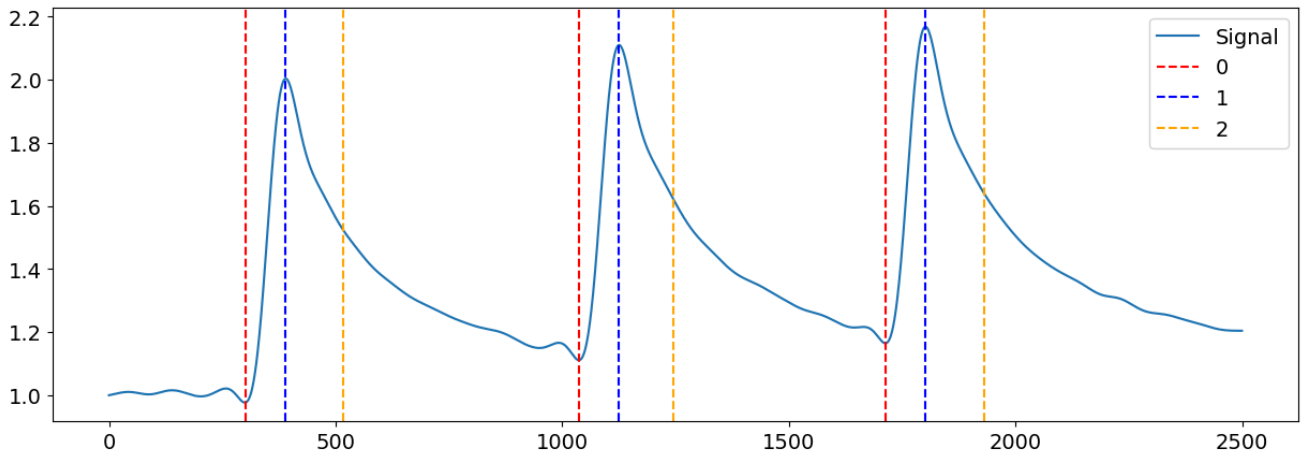
Locate Skin Conductance Response (SCR) features

The processing function does two important things for our purpose: Firstly, it cleans the signal. Secondly, it detects the location of 1) peak onsets, 2) peak amplitude, and 3) half-recovery time. Let's extract these from the output.

```
# Extract clean EDA and SCR features
cleaned = signals["EDA_Clean"]
features = [info["SCR_Onsets"], info["SCR_Peaks"], info["SCR_Recovery"]]
```

We can now visualize the location of the peak onsets, the peak amplitude, as well as the half-recovery time points in the cleaned EDA signal, respectively marked by the red dashed line, blue dashed line, and orange dashed line.

```
# Visualize SCR features in cleaned EDA signal
plot = nk.events_plot(features, cleaned, color=['red', 'blue', 'orange'])
```



Decompose EDA into Phasic and Tonic

components

We can also decompose the EDA signal into its phasic and tonic components, or more specifically, the ***Phasic Skin Conductance Response (SCR)*** and the ***Tonic Skin Conductance Level (SCL)*** respectively. The SCR represents the stimulus-dependent fast changing signal whereas the SCL is slow-changing and continuous. Separating these two signals helps to provide a more accurate estimation of the true SCR amplitude.

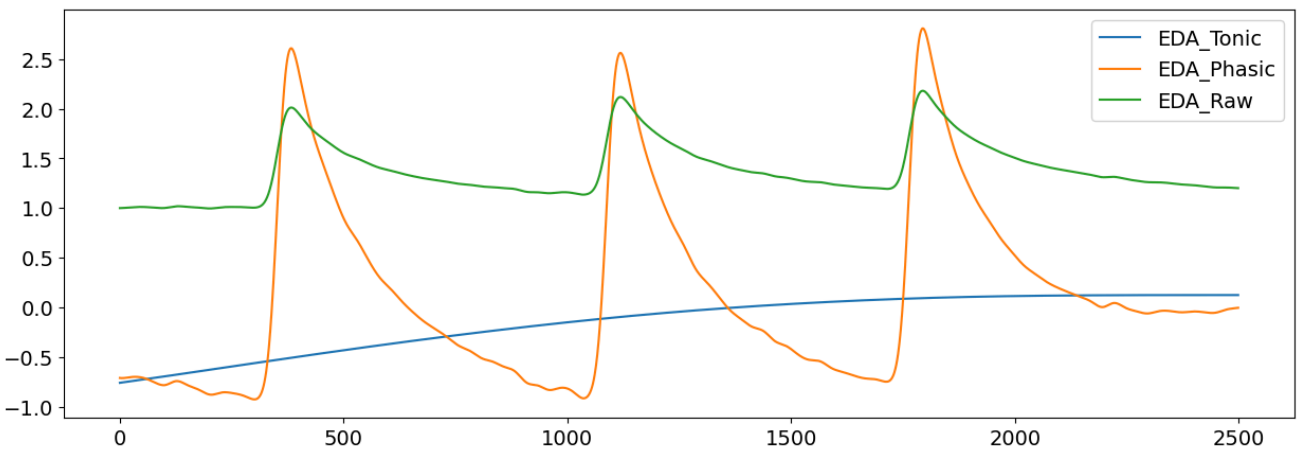
```
# Filter phasic and tonic components
data = nk.eda_phasic(nk.standardize(eda_signal), sampling_rate=250)
```

*Note: here we **standardized** the raw EDA signal before the decomposition, which can be useful in the presence of high inter-individual variations.*

We can now add the raw signal to the dataframe containing the two signals, and plot them!

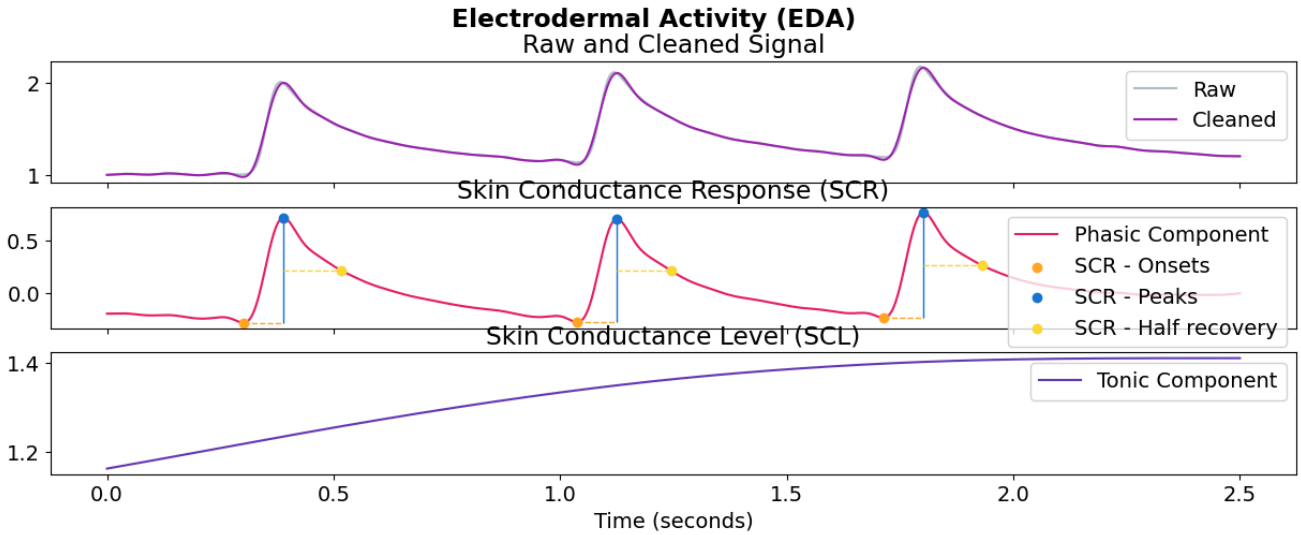
```
data["EDA_Raw"] = eda_signal  # Add raw signal
data.plot()
```

```
<Axes: >
```



Quick Plot

You can obtain all of these features by using the [eda_plot()]() function on the dataframe of processed EDA.

```
# Plot EDA signal
nk.eda_plot(signals)
```

```
C:\hostedtoolcache\windows\Python\3.13.7\x64\Lib\site-packages\neurokit2\eda\eda_pl
  warn(
```